$$\text{and} \quad y''' = (-1)(-2)(-3)x^{-4} = \frac{(-1)^3 3!}{x^4} .$$

Example 10.33

Find f'' if $f(x) = x \cos x$.

Solution

$$\begin{aligned} \text{We have, } f(x) &= x \cos x. \\ \text{Now } f'(x) &= -x \sin x + \cos x, \text{ and} \\ f''(x) &= -(x \cos x + \sin x) - \sin x \\ &= -x \cos x - 2 \sin x. \end{aligned}$$

Example 10.34

Find y'' if $x^4 + y^4 = 16$.

Solution

We have $x^4 + y^4 = 16$.

Differentiating implicitly, $4x^3 + 4y^3 y' = 0$

Solving for y' gives

$$y' = -\frac{x^3}{y^3} .$$

To find y'' we differentiate this expression for y' using the quotient rule and remembering that y is a function of x.

$$y'' = \frac{d}{dx}\left(\frac{-x^3}{y^3}\right) = \frac{-\left[y^3 \frac{d}{dx}(x^3) - x^3 \frac{d}{dx}(y^3)\right]}{(y^3)^2}$$

$$= -\frac{\left[y^3 .3x^2 - x^3(3y^2 y')\right]}{y^6}$$

$$= -\frac{3x^2 y^3 - 3x^3 y^2\left(-\frac{x^3}{y^3}\right)}{y^6}$$

$$= -\frac{3(x^2 y^4 + x^6)}{y^7} = \frac{-3x^2[x^4 + y^4]}{y^7}$$

$$= \frac{-3x^2(16)}{y^7} = \frac{-48x^2}{y^7} .$$

Example 10.35

Find the second order derivative if x and y are given by

$$x = a \cos t$$

$$y = a \sin t.$$

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